

GRAPH THEORY

(2 MARKS)

1. GATE CSE 2026 Set 2

MCQ (Single Correct Answer) Marks: +2 / -0

Consider a complete graph K_n with n vertices ($n > 4$). Note that multiple spanning trees can be constructed over K_n . Each of these spanning trees is represented as a set of edges. The Jaccard coefficient between any two sets is defined as the ratio of the size of the intersection of the two sets to the size of the union of the two sets. Which one of the following options gives the lowest possible value for the Jaccard coefficient between any two spanning trees of K_n ?

- (A) $\frac{1}{n}$
- (B) $\frac{1}{2n-3}$
- (C) 0
- (D) $\frac{1}{n-1}$

2. GATE CSE 2026 Set 1

MCQ (Single Correct Answer) Marks: +2 / -0

Let $G(V, E)$ be a simple, undirected graph. A vertex cover of G is a subset $V' \subseteq V$ such that for every $(u, v) \in E$, $u \in V'$ or $v \in V'$. Let the size of the smallest vertex cover in G be k . Let S be any vertex cover of size k .

For a vertex $v \in V$, which of the following constraints will always ensure that $v \in S$?

- (A) The degree of v is at least $k + 1$
- (B) The vertex v is on a path of length $k + 1$
- (C) The vertex v is on a cycle of length $k + 1$
- (D) The vertex v is a part of a clique of size k

3. GATE CSE 2026 Set 1

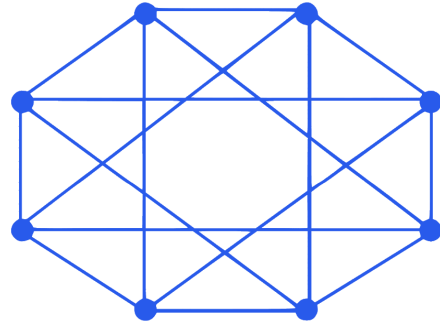
Numerical Marks: +2 / -0

Let G be an undirected graph, which is a path on 8 vertices. The number of matchings in G is _____ (answer in integer)

4. GATE CSE 2024 Set 2

Numerical Marks: +2 / -0

The chromatic number of a graph is the minimum number of colours used in a proper colouring of the graph. The chromatic number of the following graph is _____



5. GATE CSE 2024 Set 1

MCQ (More than One Correct Answer) Marks: +2 / -0

The chromatic number of a graph is the minimum number of colours used in a proper colouring of the graph. Let G be any graph with n vertices and chromatic number k . Which of the following statements is/are always TRUE?

- (A) G contains a complete subgraph with k vertices
- (B) G contains an independent set of size at least n/k
- (C) G contains at least $k(k-1)/2$ edges
- (D) G contains a vertex of degree at least k

6. GATE CSE 2024 Set 1

Numerical Marks: +2 / -0

The number of edges present in the forest generated by the DFS traversal of an undirected graph G with 100 vertices is 40. The number of connected components in G is _____

7. GATE CSE 2023

MCQ (More than One Correct Answer) Marks: +2 / -0

Let G be a simple, finite, undirected graph with vertex set $\{v_1, \dots, v_n\}$. Let $\Delta(G)$ denote the maximum degree of G and let $N = \{1, 2, \dots\}$ denote the set of all possible colors. Color the vertices of G using the following greedy strategy: for $i = 1, \dots, n$

$\text{color}(v_i) \leftarrow \min\{j \in N : \text{no neighbour of } v_i \text{ is colored } j\}$

Which of the following statements is/are TRUE?

- (A) This procedure results in a proper vertex coloring of G .
- (B) The number of colors used is at most $\Delta(G) + 1$.
- (C) The number of colors used is at most $\Delta(G)$.
- (D) The number of colors used is equal to the chromatic number of G .

8. GATE CSE 2023

Numerical Marks: +2 / -0

Let $U = \{1, 2, 3\}$. Let 2^U denote the powerset of U . Consider an undirected graph G whose vertex set is 2^U . For any $A, B \in 2^U$, (A, B) is an edge in G if and only if (i) $A \neq B$, and (ii) either $A \supsetneq B$ or $B \supsetneq A$. For any vertex A in G , the set of all possible orderings in which the vertices of G can be visited in a Breadth First Search (BFS) starting from A is denoted by $B(A)$.

If ϕ denotes the empty set, then the cardinality of $B(\phi)$ is _____

9. GATE CSE 2022

MCQ (Single Correct Answer) Marks: +2 / -0.67

Consider a simple undirected unweighted graph with at least three vertices. If A is the adjacency matrix of the graph, then the number of 3-cycles in the graph is given by the trace of

- (A) A^3
- (B) A^3 divided by 2
- (C) A^3 divided by 3
- (D) A^3 divided by 6

10. GATE CSE 2022

MCQ (More than One Correct Answer) Marks: +2 / -0

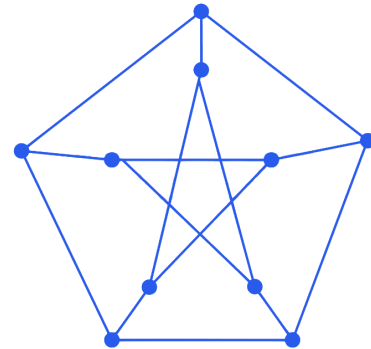
Consider a simple undirected weighted graph G , all of whose edge weights are distinct. Which of the following statements about the minimum spanning trees of G is/are TRUE?

- (A) The edge with the second smallest weight is always part of any minimum spanning tree of G .
- (B) One or both of the edges with the third smallest and the fourth smallest weights are part of any minimum spanning tree of G .
- (C) Suppose $S \subseteq V$ be such that $S \neq \phi$ and $S \neq V$. Consider the edge with the minimum weight such that one of its vertices is in S and the other in $V \setminus S$. Such an edge will always be part of any minimum spanning tree of G .
- (D) G can have multiple minimum spanning trees.

11. GATE CSE 2022

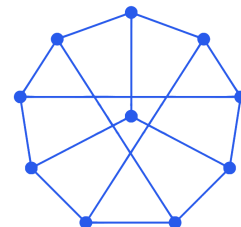
MCQ (More than One Correct Answer) Marks: +2 / -0

The following simple undirected graph is referred to as the Peterson graph.



Which of the following statements is/are TRUE?

- (A) The chromatic number of the graph is 3.
- (B) The graph has a Hamiltonian path.
- (C) The following graph is isomorphic to the Peterson graph.



- (D) The size of the largest independent set of the given graph is 3. (A subset of vertices of a graph form an independent set if no two vertices of the subset are adjacent.)

12. GATE CSE 2022

MCQ (Single Correct Answer) Marks: +2 / -0.67

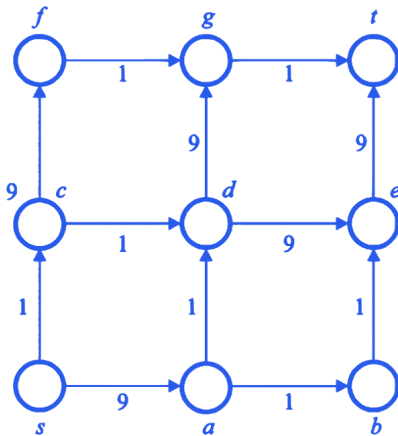
Which of the properties hold for the adjacency matrix A of a simple undirected unweighted graph having n vertices?

- (A) The diagonal entries of A^2 are the degrees of the vertices of the graph.
- (B) If the graph is connected, then none of the entries of $A^{n-1} + I_n$ can be zero.
- (C) If the sum of all the elements of A is at most $2(n-1)$, then the graph must be acyclic.
- (D) If there is at least a 1 in each of A 's rows and columns, then the graph must be connected.

13. GATE CSE 2021 Set 2

Numerical Marks: +2 / -0

In a directed acyclic graph with a source vertex s , the quality-score of a directed path is defined to be the product of the weights of the edges on the path. Further, for a vertex v other than s , the quality-score of v is defined to be the maximum among the quality-scores of all the paths from s to v . The quality-score of s is assumed to be 1.



The sum of the quality-scores of all the vertices in the graph shown above is _____

14. GATE CSE 2021 Set 1

MCQ (More than One Correct Answer) Marks: +2 / -0

An articulation point in a connected graph is a vertex such that removing the vertex and its incident edges disconnects the graph into two or more connected components.

Let T be a DFS tree obtained by doing DFS in a connected undirected graph G . Which of the following option is/are correct?

- (A) If u is an articulation point in G such that x is an ancestor of u in T and y is a descendent of u in T , then all paths from x to y in G must pass through u .
- (B) Root of T is an articulation point in G if and only if it has 2 or more children.
- (C) Root of T can never be an articulation point in G .
- (D) A leaf of T can be an articulation point in G .

15. GATE CSE 2021 Set 1

MCQ (Single Correct Answer) Marks: +2 / -0.67

Let G be a group order 6, and H be a subgroup of G such that $1 < |H| < 6$. Which one of the following options is correct?

- (A) G is always cyclic, but H may not be cyclic.
- (B) G may not be cyclic, but H is always cyclic.
- (C) Both G and H are always cyclic.
- (D) Both G and H may not be cyclic.

16. GATE CSE 2021 Set 1

MCQ (Single Correct Answer) Marks: +2 / -0.67

Let $G = (V, E)$ be an undirected unweighted connected graph. The diameter of G is defined as:

$$\text{diam}(G) = \max_{u, v \in V} \{\text{the length of shortest path between } u \text{ and } v\}$$

Let M be the adjacency matrix of G .

Define graph G_2 on the same set of vertices with adjacency matrix N , where

$$N_{ij} = \begin{cases} 1 & \text{if } M_{ij} > 0 \text{ or } P_{ij} > 0, \text{ where } P = M^2 \\ 0, & \text{otherwise} \end{cases}$$

Which one of the following statements is true?

- (A) $\text{diam}(G) < \text{diam}(G_2) \leq \text{diam}(G)$
- (B) $\lceil \text{diam}(G)/2 \rceil < \text{diam}(G_2) < \text{diam}(G)$
- (C) $\text{diam}(G_2) \leq \lceil \text{diam}(G)/2 \rceil$
- (D) $\text{diam}(G_2) = \text{diam}(G)$

17. GATE CSE 2020

Numerical Marks: +2 / -0

Graph G is obtained by adding vertex s to $K_{3,4}$ and making s adjacent to every vertex of $K_{3,4}$. The minimum number of colours required to edge-colour G is _____.

18. GATE CSE 2019

MCQ (Single Correct Answer) Marks: +2 / -0.67

Let G be any connected, weighted, undirected graph.

I. G has a unique minimum spanning tree, if no two edges of G have the same weight.

II. G has a unique minimum spanning tree, if, for every cut of G , there is a unique minimum-weight edge crossing the cut.

Which of the above two statements is/are TRUE?

- (A) I only
- (B) II only
- (C) Both I and II
- (D) Neither I nor II

19. GATE CSE 2015 Set 2

MCQ (Single Correct Answer) Marks: +2 / -0.6

In a connected graph, bridge is an edge whose removal disconnects a graph. Which one of the following statements is true?

- (A) A tree has no bridges
- (B) A bridge cannot be part of a simple cycle
- (C) Every edge of a clique with size ≥ 3 is a bridge (A clique is any complete sub-graph of a graph)
- (D) A graph with bridges cannot have a cycle

20. GATE CSE 2015 Set 2

MCQ (Single Correct Answer) Marks: +2 / -0.6

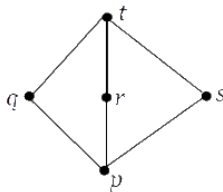
A graph is self-complementary if it is isomorphic to its complement. For all self-complementary graphs on n vertices, n is

- (A) A multiple of 4
- (B) Even
- (C) Odd
- (D) Congruent to 0 mod 4, or, 1 mod 4.

21. GATE CSE 2015 Set 1

MCQ (Single Correct Answer) Marks: +2 / -0.6

Suppose $L = \{p, q, r, s, t\}$ is a lattice represented by the following Hasse diagram:



For any $x, y \in L$, not necessarily distinct, $x \vee y$ and $x \wedge y$ are join and meet of x, y , respectively. Let $L^3 = \{(x, y, z) : x, y, z \in L\}$ be the set of all ordered triplets of the elements of L . Let p_r be the probability that an element $(x, y, z) \in L^3$ chosen equiprobably satisfies $x \vee (y \wedge z) = (x \vee y) \wedge (x \vee z)$. Then

- (A) $p_r = 0$
- (B) $p_r = 1$
- (C) $0 < p_r \leq \frac{1}{5}$
- (D) $\frac{1}{5} < p_r < 1$

22. GATE CSE 2015 Set 1

Numerical Marks: +2 / -0

Let G be a connected planar graph with 10 vertices. If the number of edges on each face is three, then the number of edges in G is _____.

23. GATE CSE 2014 Set 2

Numerical Marks: +2 / -0

A cycle on n vertices is isomorphic to its complement. The value of n is _____.

24. GATE CSE 2014 Set 1

Numerical Marks: +2 / -0

Consider an undirectional graph G where self-loops are not allowed. The vertex set of G is $\{(i, j) : 1 \leq i \leq 12, 1 \leq j \leq 12\}$. There is an edge between (a, b) and (c, d) if $|a - c| \leq 1$ and $|b - d| \leq 1$. The number of edges in this graph is _____.

25. GATE CSE 2014 Set 1

MCQ (Single Correct Answer) Marks: +2 / -0.6

An ordered n -tuple $(d_1, d_2, \dots, d_n)$ with $d_1 \geq d_2 \geq \dots \geq d_n$ is called graphic if there exists a simple undirected graph with n vertices having degrees $d_1, d_2, \dots, d_n$ respectively. Which of the following 6-tuples is NOT graphic?

- (A) (1, 1, 1, 1, 1, 1)
- (B) (2, 2, 2, 2, 2, 2)
- (C) (3, 3, 3, 1, 0, 0)
- (D) (3, 2, 1, 1, 1, 0)

26. GATE CSE 2014 Set 3

MCQ (Single Correct Answer) Marks: +2 / -0.6

If G is a forest with n vertices and k connected components, how many edges does G have?

- (A) $\lfloor n/k \rfloor$
- (B) $\lceil n/k \rceil$
- (C) $n - k$
- (D) $n - k + 1$